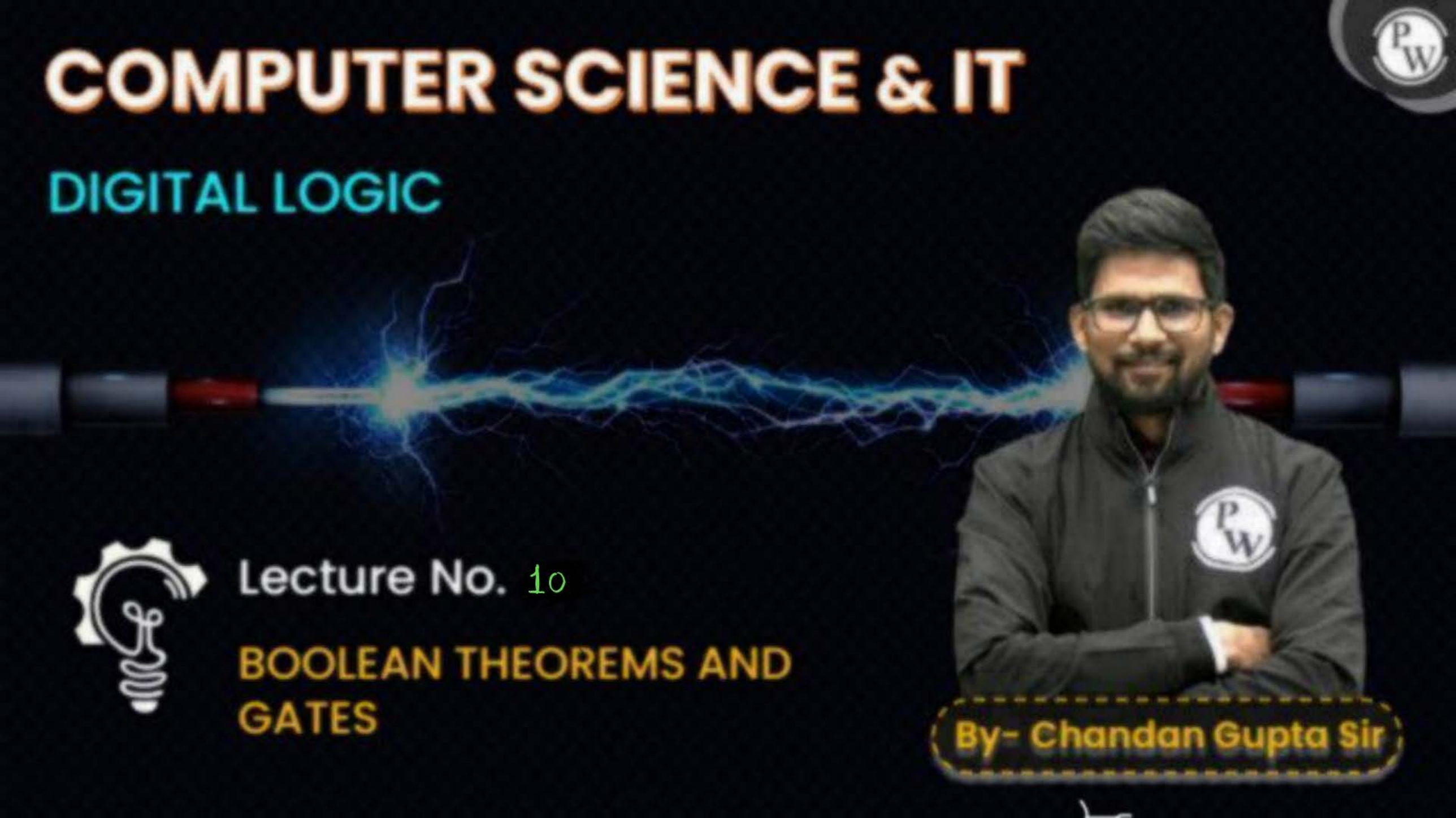
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 10

BOOLEAN THEOREMS AND GATES

By- Chandan Gupta Sir



NAND gate



- NOR gate

Q. $f(A,B,C) = \bar{A}\bar{B} + A\bar{B} + BC$

Minimum no. of 2-i/p NAND gate required to implement $f(A,B,C)$ ____6____.

$$\bar{f} = \left(\overline{\bar{A}\cdot\bar{B}} \quad \overline{A\bar{B}}\right) \quad \overline{BC}$$

$\overline{\bar{A}\cdot\bar{B}} \rightarrow (A+B)(\bar{A}+\bar{B})$ → P (3 NAND)

$\overline{A\bar{B}}$ → 1 NAND → Q

$\overline{BC}$ → 1 NAND → R

$$f = \overline{P\cdot Q\cdot R} \longrightarrow 3 \text{ NAND}$$

$$f(A,B,C) = A \odot B + BC$$

$$\bar{f} = \overline{A \odot B + BC} = \overline{A \odot B} \cdot \overline{BC}$$

$$\bar{f} = (A \oplus B) \cdot \overline{BC}$$

$(A \oplus B)$ → 4 NAND → W

$\overline{BC}$ → 1 NAND (Z)

$$\bar{f} = W \cdot Z$$

$$f = \overline{W \cdot Z} \longrightarrow 1 \text{ NAND}$$

Question

$y = A\bar{B} + BC$

(H.W)

Minimum no. of 2-input NAND gates required to implement y _____

Ans. 4.

Question

$y = \bar{A}BC$

To implement y, minimum no. of 2-input NAND gates required is _____

Ans. 5

H.W

H.W.

Minimum no. 2-i/p NAND gate required to implement

1. $f(A,B) = A\bar{B} \oplus \bar{A}\bar{B}$ ⟶ Ans. 1

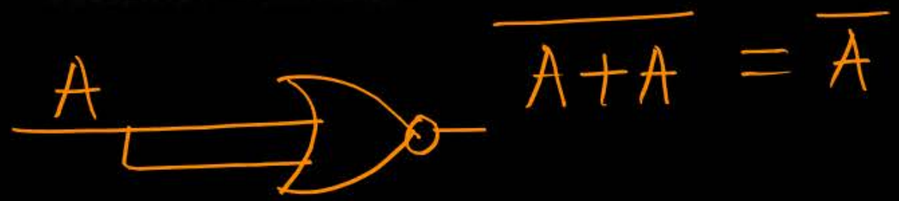
2. $f(A,B,D) = \overline{AB} + \bar{B} + \bar{D}$ ⟶ Ans. 3

3. $f(A,B,C) = AB \oplus BC \oplus ABC$ ⟶ Ans. 3

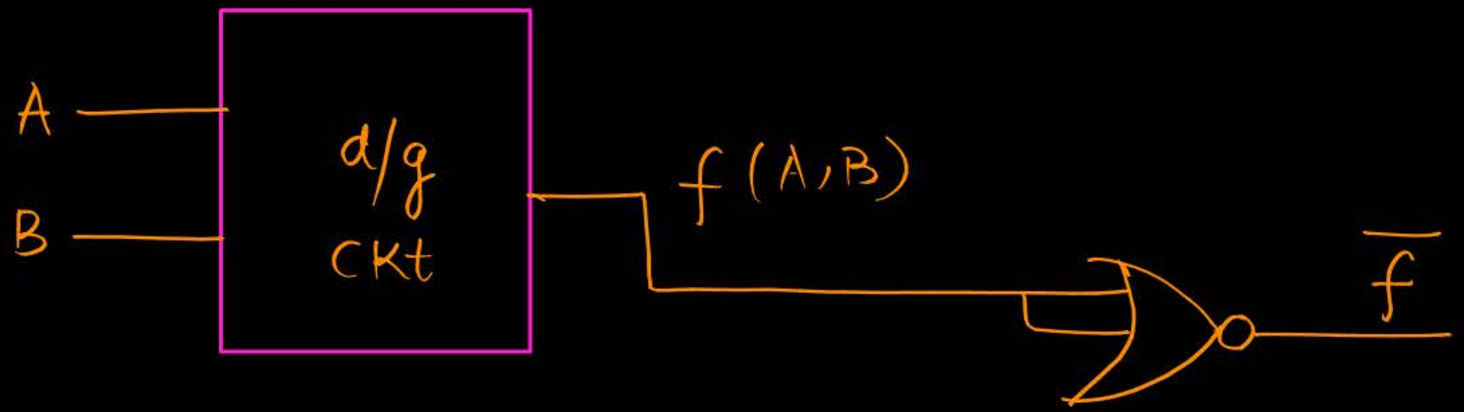
4. $f(A,B,C) = \bar{A}B + A\bar{B} + BC$ ⟶ Ans. 7 ✓

NOR GATE as Universal Circuit

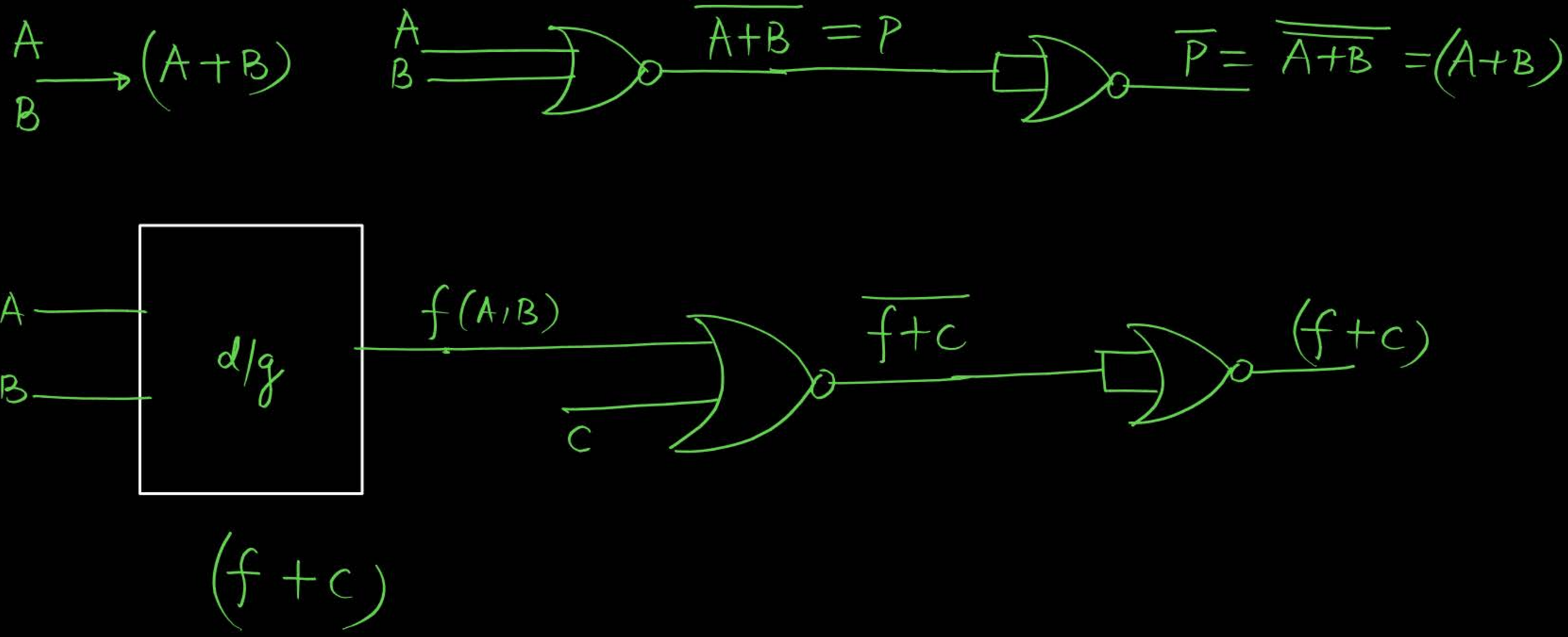
- NOT GATE



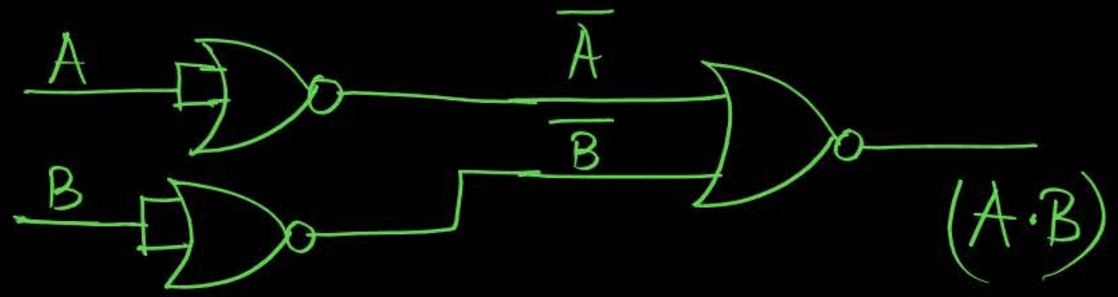
$\overline{A+A} = \overline{A}$

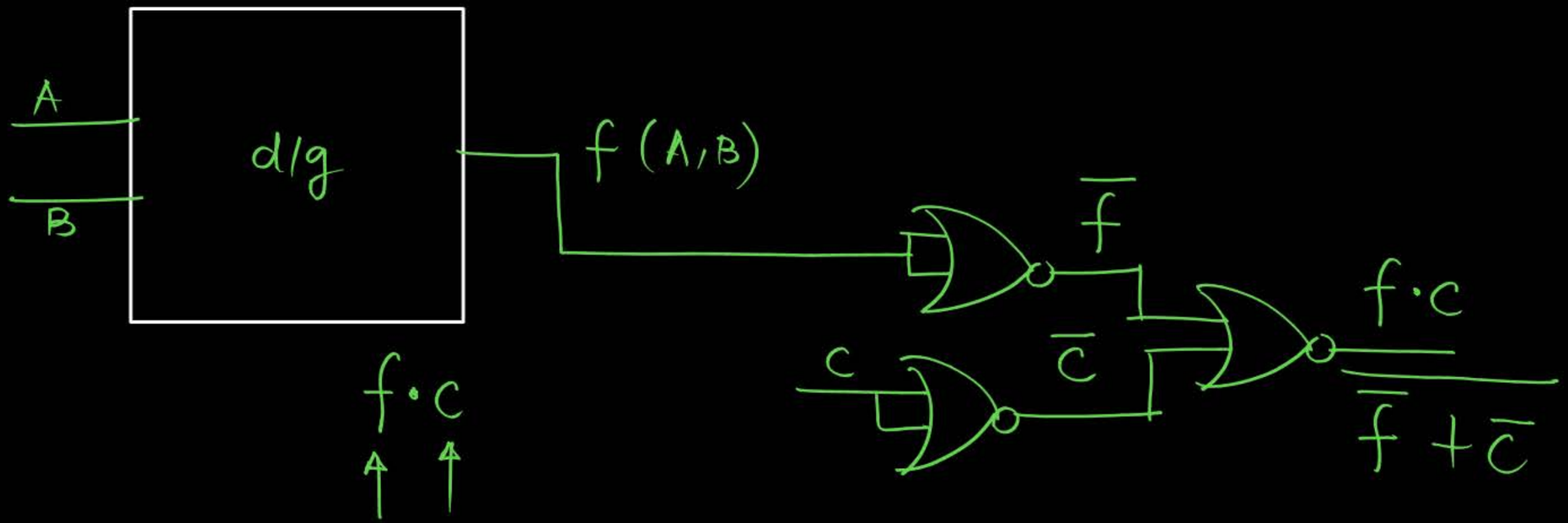


- OR GATE

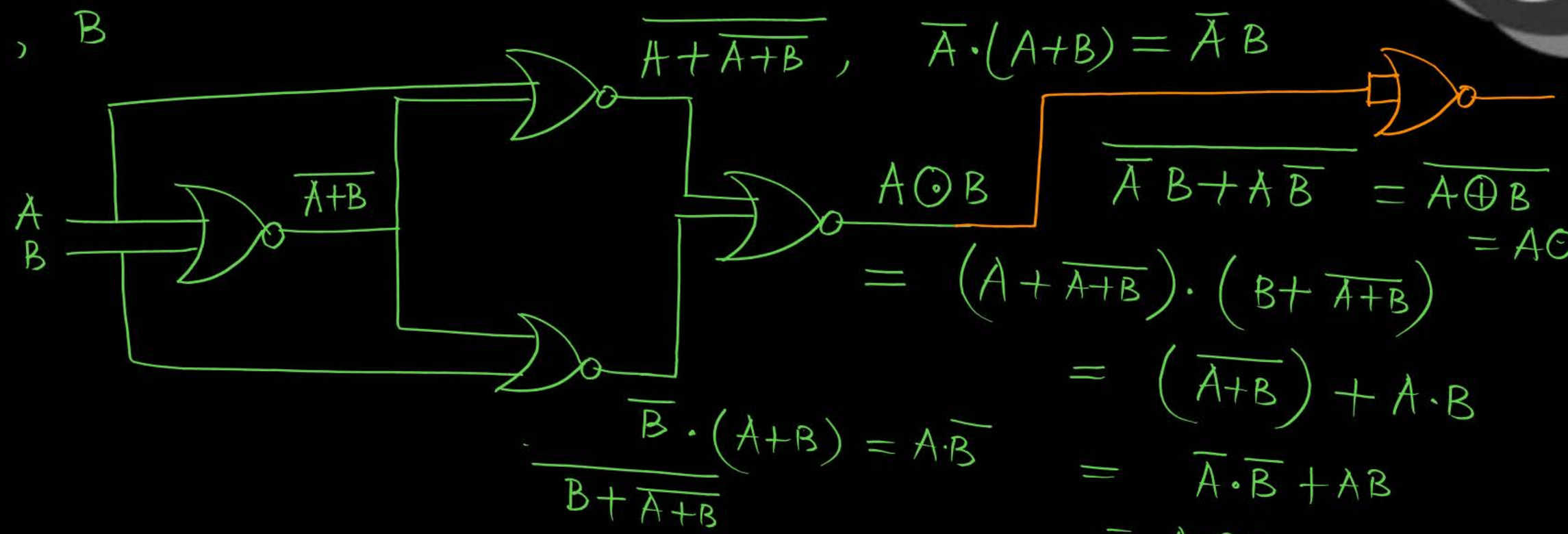

A
B
(A+B)
A
B
$\overline{A+B} = P$
$\overline{P} = \overline{\overline{A+B}} = (A+B)$
A
B
d/g
f(A,B)
C
$\overline{f+C}$
(f+C)
(f+C)

- AND GATE $A \cdot B$





[XOR and XNOR GATE using NOR GATE]



$$= \left(A+\overline{A+B}\right)\cdot\left(B+\overline{A+B}\right)$$

$$= \left(\overline{A+B}\right)+A\cdot B$$

$$= \overline{A}\cdot\overline{B}+AB$$

$$= A\odot B$$

GATES	No. of NAND	No. of NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

- $f(A,B,C) = (A+B)\cdot(B+C)$

Minimum no of 2 i/p NOR gate required to implement $f(A,B,C)$ ___3___.

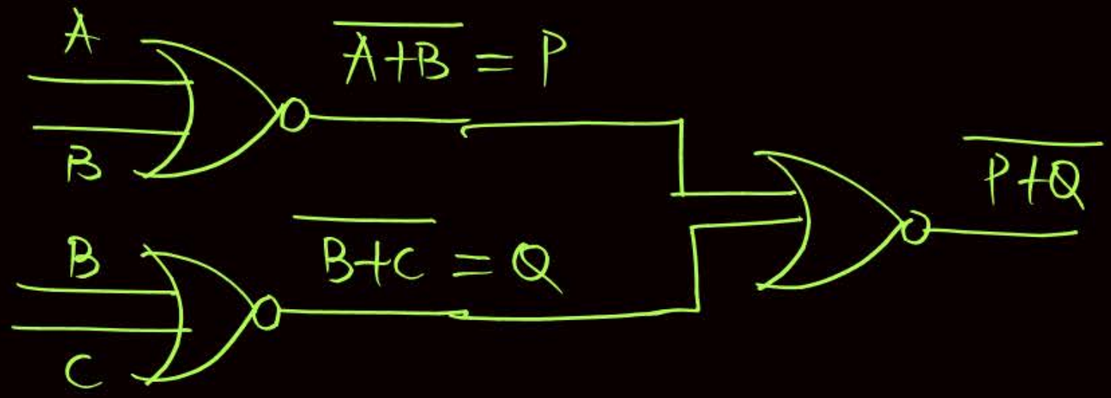
$$\overline{f} = \overline{(A+B)\cdot(B+C)} = \overline{A+B} + \overline{(B+C)}$$

$\overline{A+B}$ → 1 NOR (P)

$\overline{(B+C)}$ → 1 NOR (Q)

$\overline{f} = (P+Q)$

$f = \overline{P+Q}$ → 1 NOR



NOT	1 NOR
$A+B$	OR → 2 NOR
$\overline{A+B}$	NOR → 1 NOR
$A\cdot B$	AND → 3 NOR
$\overline{A\cdot B}$	NAND → 4 NOR

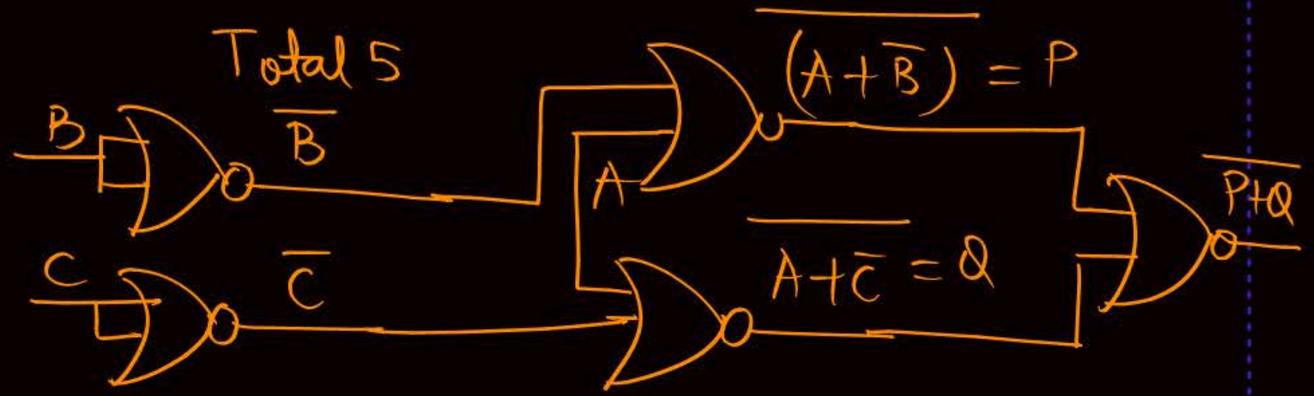
Q. $f(A,B,C) = (A+\bar{B})\cdot(A+\bar{C})$

$$\bar{f} = \overline{(A+\bar{B})} + \overline{(A+\bar{C})}$$

2NOR (P) 2NUR (Q)

$\bar{f} = P+Q$

$f = \overline{P+Q}$ → INOR



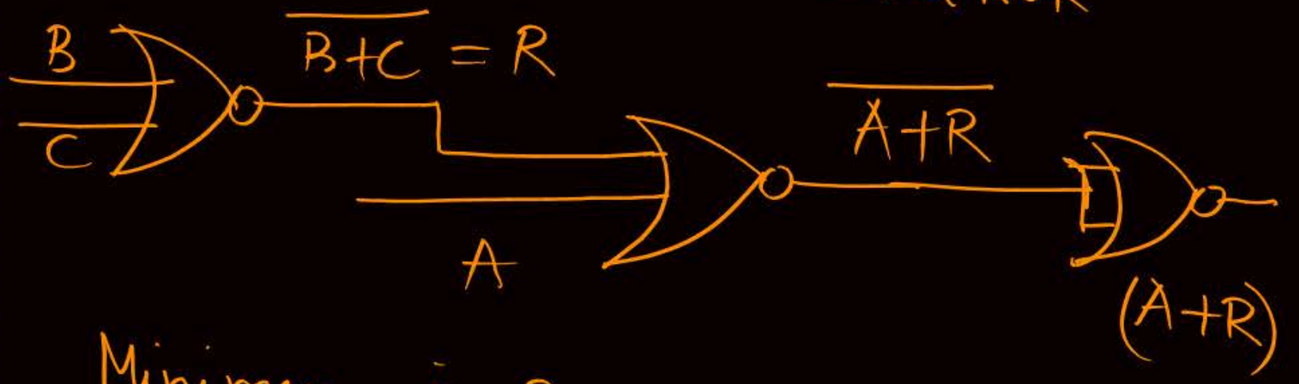
$(A+\bar{B})(A+\bar{C}) = A + \bar{B}\cdot\bar{C}$

$= A + \overline{(B+C)}$

↓ INOR

$= A + R$

↓ 2 NOR



Minimum is 3.

Q. $f(A,B) = (\overline{A}+B)\cdot(A+\overline{B}) = \overline{A}\,\overline{B} + AB = \overline{A+B} + AB$

$= (R+(A\cdot B)) = (R+A)\cdot(R+B)$

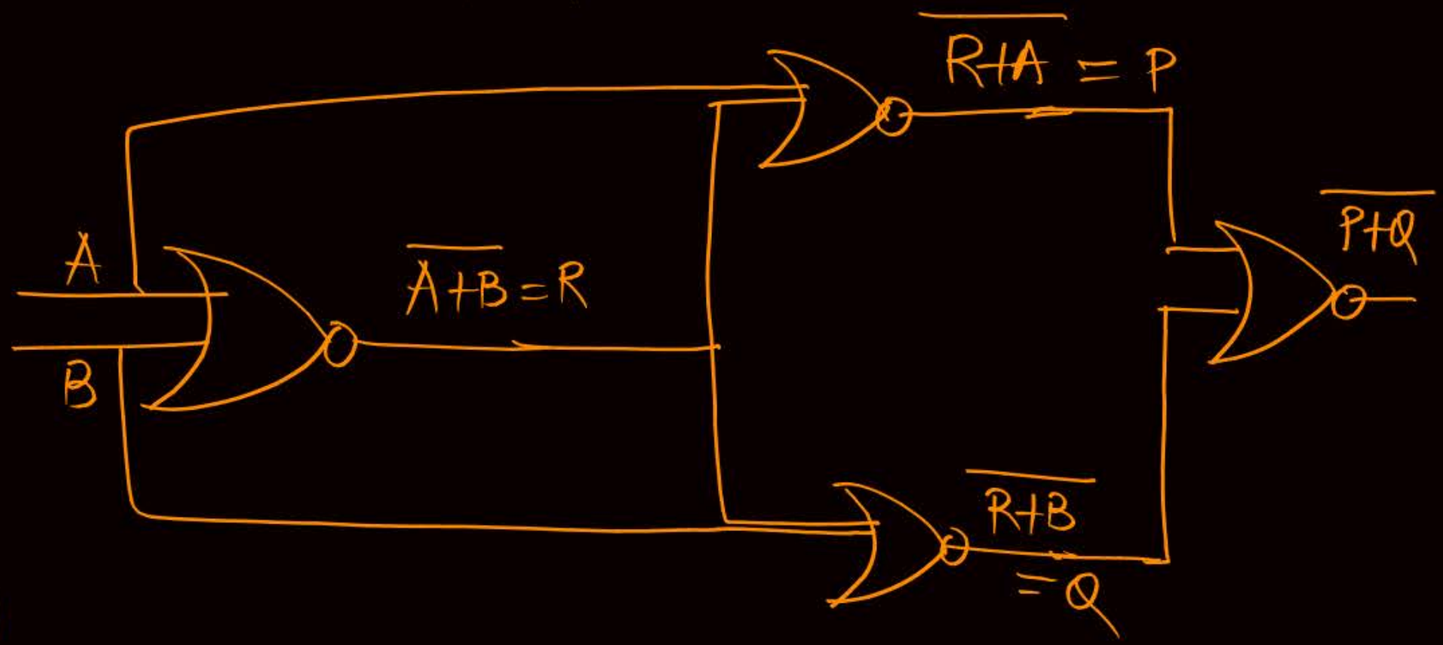
INOR

$\overline{f} = \overline{R+A} + \overline{R+B}$

INOR (P) INOR Q

$\overline{f} = P+Q$

$f = \overline{P+Q}$ → INOR



- $f(A,B,C) = \overline{A}\,\overline{B} + \overline{B}\,\overline{C}$

Minimum no of 2 i/p NOR gate to implement $f(A,B,C)$ $\frac{4}{\overline{B}\,[\overline{A}\,\overline{C}]}$.

$$f = \overline{A}\,\overline{B} + \overline{B}\,\overline{C}$$

$$= \overline{A+B} + \overline{B+C}$$

1 NOR P 1 NOR Q

$$f = (P+Q)$$

→ 2 NOR

$$f = (\overline{B}).[\overline{A}+\overline{C}]$$

$$\overline{f} = B + \overline{\overline{A}+\overline{C}}$$

3 NOR

$$\overline{f} = B+R$$

R

$$f = \overline{B+R}$$ — 1 NOR

- $f(A,B,C) = (A + BC)$

Minimum no. of 2 i/p NAND gate required to implement $f(A,B,C)$ — 3.

Minimum no. of 2 i/p NOR gate required to implement $f(A,B,C)$ — 3.

$$\overline{f} = \overline{A} \cdot \overline{BC}$$

$\overline{A}$ → 1 NAND P, $\overline{BC}$ → 1 NAND Q

$$\overline{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \longrightarrow 1 \text{ NAND}$$

$$f = (A + BC) = (A + B) \cdot (A + C)$$

$$\overline{f} = \overline{A+B} + \overline{A+C}$$

$\overline{A+B}$ → 1 NOR W, $\overline{A+C}$ → 1 NOR Z

$$\overline{f} = W + Z$$

$$f = \overline{W + Z} \longrightarrow 1 \text{ NOR}$$

- $f(A,B,C,D) = AB + BC + AD + CD$

Minimum no. of 2-i/p NOR gate required to implement $f(A,B,C,D)$ 3.

$= B(A+C) + D(A+C)$ ✓✓ $= B \cdot R + D \cdot R$

$f = (A+C) \cdot (B+D) \Rightarrow$

$\overline{f} = \overline{A+C} + \overline{B+D}$

1 NOR P, 1 NOR Q

$\overline{f} = P + Q$

$f = \overline{P+Q}$ → 1 NOR

$B \cdot R + D \cdot R$ → 3 NAND

2-i/p NAND 6

$f = B \cdot R + D \cdot R$

$\overline{f} = \overline{B \cdot R} \quad \overline{D \cdot R}$

$\overline{B \cdot R}$ → 1 NAND X

$\overline{D \cdot R}$ → 1 NAND Y

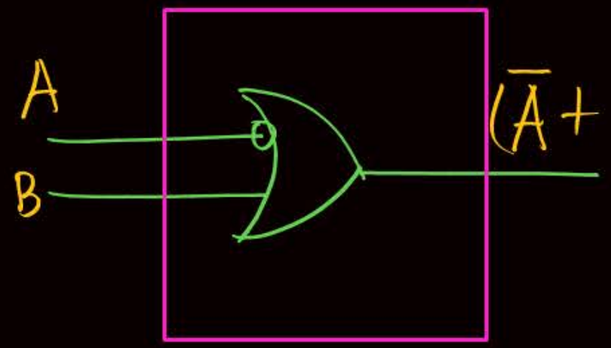
$f = \overline{X \cdot Y}$ → 1 NAND

- AND + NOT

$$\overline{\overline{A} \cdot \overline{B}} = (A + B)$$

OR + NOT

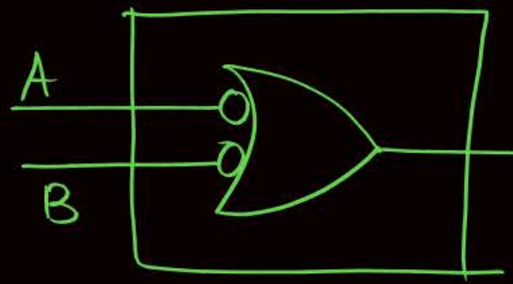
$$\overline{\overline{A} + \overline{B}} = A \cdot B$$



$(\bar{A}+B)$

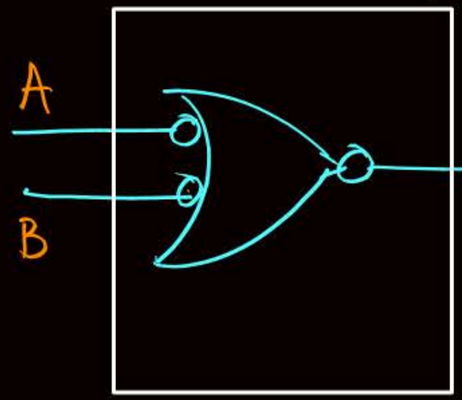
Universal

(NOT + OR)



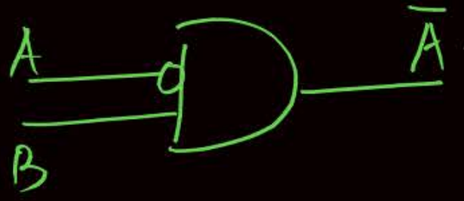
$\bar{A}+\bar{B} = \overline{A \cdot B}$

Universal



$\overline{\bar{A}+\bar{B}} = A \cdot B$

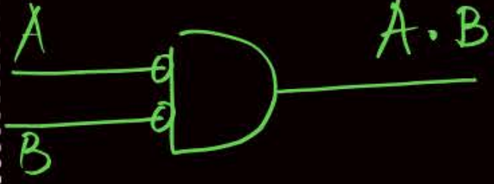
AND → Non universal



$\bar{A} \cdot B$ H.W.

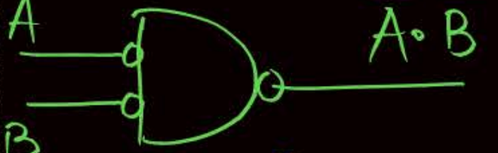
AND + NOT

Universal



$\bar{A} \cdot \bar{B} = \overline{A+B}$

Universal

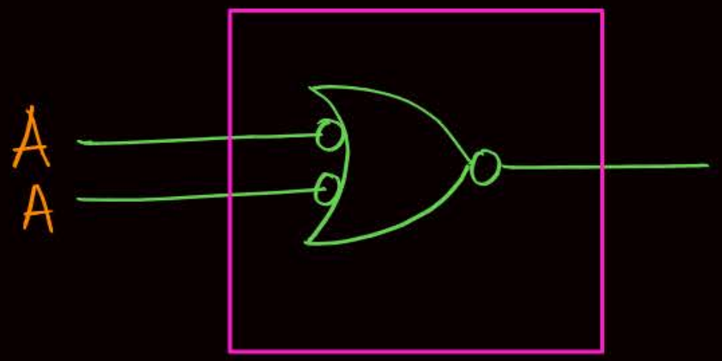


$\overline{\bar{A} \cdot \bar{B}} = (A+B)$

H.W

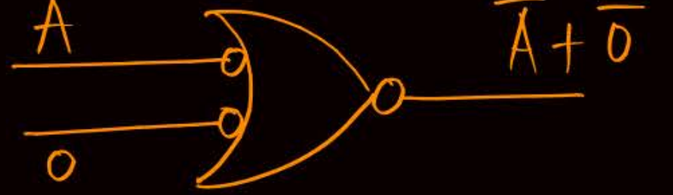
OR

↓ non universal

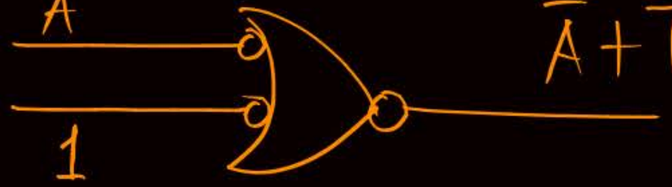


$\overline{\overline{A}\cdot + \overline{A}} = A$

A, 0, 1

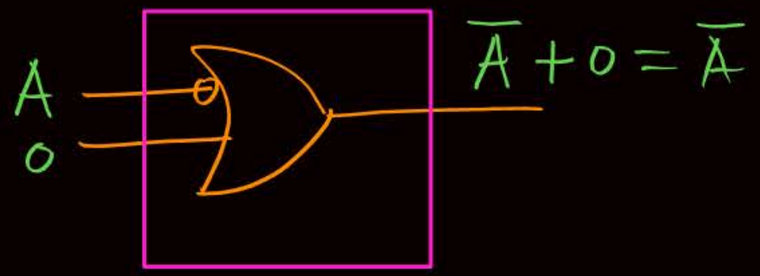


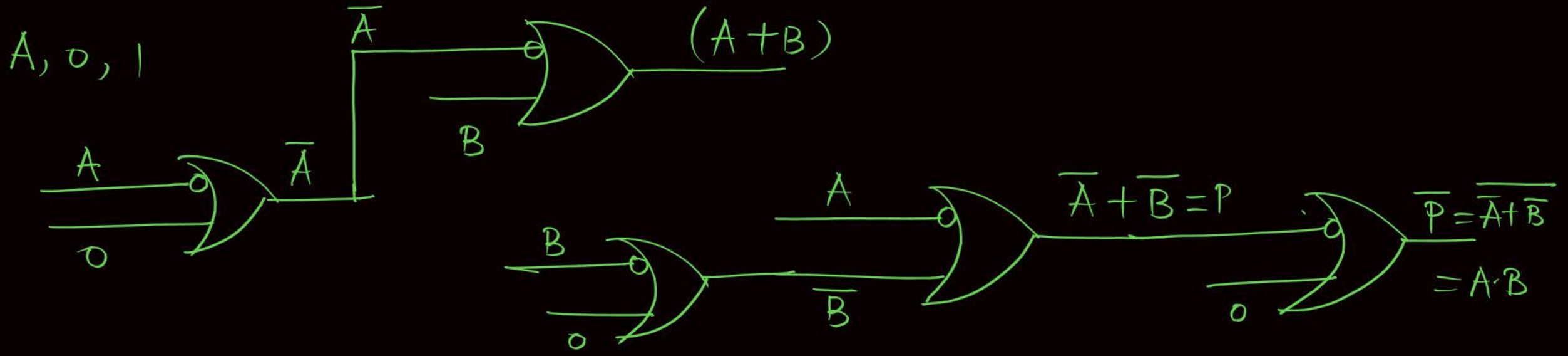
$\overline{\overline{A} + \overline{0}} = 0$



$\overline{\overline{A} + \overline{1}} = A$

NOT operation is not implemented therefore it is non-universal gate.


A
0
$\bar{A}+0=\bar{A}$


A, 0, 1
$\bar{A}$
$(A+B)$
B
A
$\bar{A}$
0
A
$\bar{A}+\bar{B}=P$
B
$\bar{P}=\overline{\bar{A}+\bar{B}}$
$=A\cdot B$
B
0
$\bar{B}$
0

Question

MSQ

Which of the following circuit can work as universal gate ?

(a) 

(b) 

H.W

(c) 

(d) 



2 Minute Summary

- NOR gate

Thank you

GW

Soldiers !

